

DIGITAL IMAGE PROCESSING

LECTURE # 10

COLOR IMAGE PROCESSING-I

Contact Information

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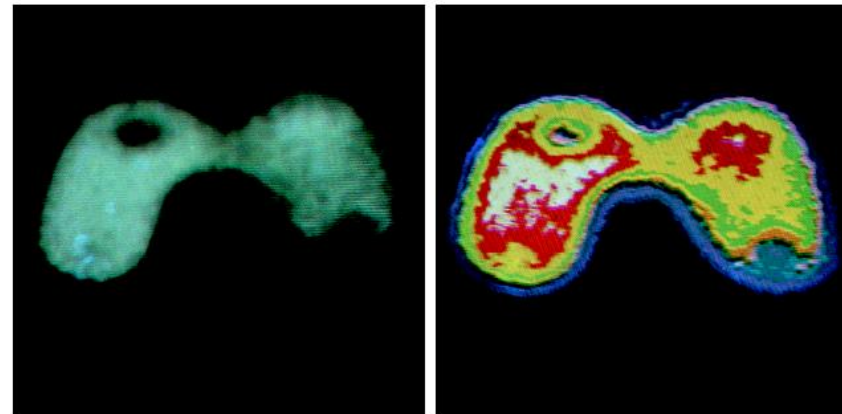
Topics to Cover

- Color Image Processing
- Color Spectrum
- Color Fundamentals
- Human Perception
- Primary color of lights vs primary color of pigments
- Chromaticity Diagram
- Color Models
 - ❑ RGB
 - ❑ CMY
 - ❑ CMYK
 - ❑ HIS
 - ❑ YIQ
 - ❑ YUV

Color Image Processing

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- ❑ The use of color Image Processing is motivated by two principal factors:
 - ✓ Color is a powerful descriptor
 - Simplifies Object identification and extraction from the scene eg. Face detection using skin colors
 - ✓ Humans can distinguish between thousands of color shades and intensities compared to about only two dozen shades of gray



Color Image Processing

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- ❑ In **Full Color Processing** the image in question typically are acquired with a Full-Color sensor e.g., Color TV camera or Color Scanner
- ❑ In **Pseudo-color Processing** the problem is of assigning a color to a particular monochrome intensity or a range of intensities
- ❑ **Pseudo-color Processing** is quite useful to assign colors for human interpretation

Color Image Fundamentals

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- ❑ Perception and interpretation of color is a physio-psychological phenomenon that is not fully understood.
- ❑ The colors that human perceive in an object are determined by the nature of the light reflected from that object.
 - ❑ Green objects reflect light with wavelengths in 500-570nm while absorbing most of the energy at other wavelengths.
- ❑ Characterization of light is central to the science of color.
 - ❑ Achromatic light
 - ❑ Chromatic light

Color Spectrum

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- In 1666 Sir Isaac Newton discovered that when a beam of light passes through a glass prism, the emerging beam of light is not white but consists instead of a continuous spectrum of colors ranging from violet at one end to red at the other, as shown in Figure 6.1

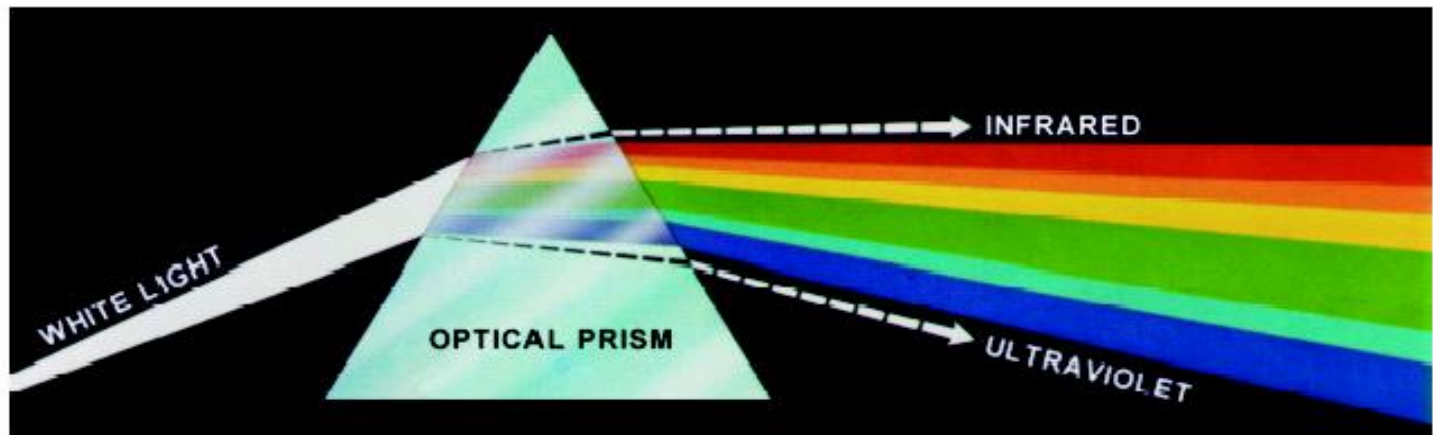


FIGURE 6.1 Color spectrum seen by passing white light through a prism. (Courtesy of the General Electric Co., Lamp Business Division.)

Chromatic Light

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- ❑ Chromatic light spans the EM spectrum from 400nm to 700nm.
- ❑ 3 basic quantities to describe the quality of a chromatic light source:
 - ❑ Radiance
 - ❑ Luminance
 - ❑ Brightness

Human Perception

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- ❑ Detailed experimental evidences has established that the 6 to 7 million cones in the human eye can be divided into three principal sensing categories, corresponding roughly to red, green and blue
- ❑ Approximately 65% of all cones are sensitive to Red Light, 33% are sensitive to Green Light and about 2% are sensitive to Blue Light

Human Perception

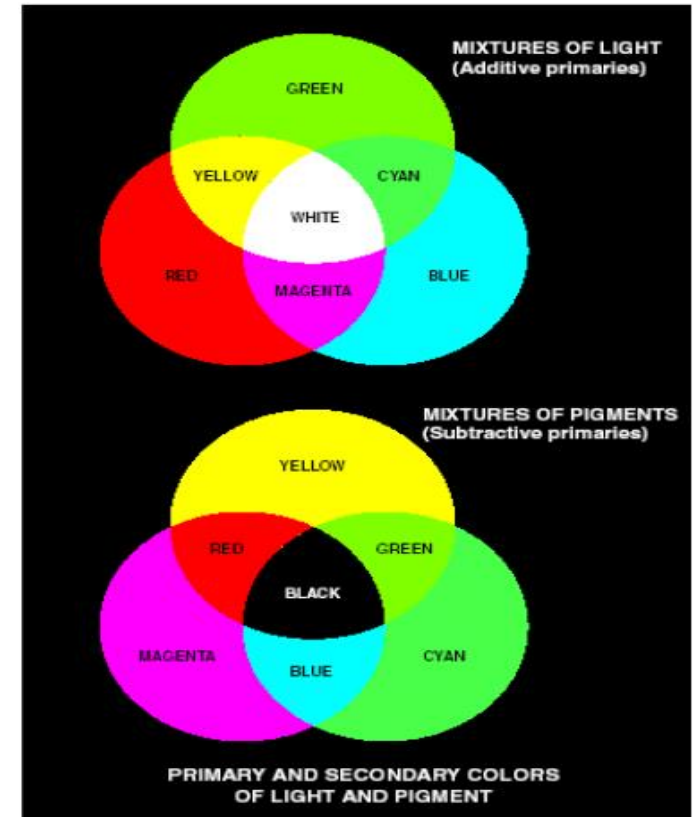
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- ❑ Due to these absorption characteristic of Human Eye colors are seen as variable combinations of the so-called 'Primary Colors' Red, Green and Blue
- ❑ The primary colors can be added to produce secondary colors of Light
 - ❑ Magenta (Red+Blue)
 - ❑ Cyan (Green+Blue)
 - ❑ Yellow (Red+Green)

Color Fundamentals

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- ❑ In 1931, CIE (International Commission on Illumination) defines specific wavelength values to the primary colors
 - ✓ $B = 435.8 \text{ nm}$, $G = 546.1 \text{ nm}$, $R = 700 \text{ nm}$
 - ✓ However, we know that no single color may be called red, green, or blue



a
b

FIGURE 6.4 Primary and secondary colors of light and pigments. (Courtesy of the General Electric Co., Lamp Business Division.)

Color Fundamentals

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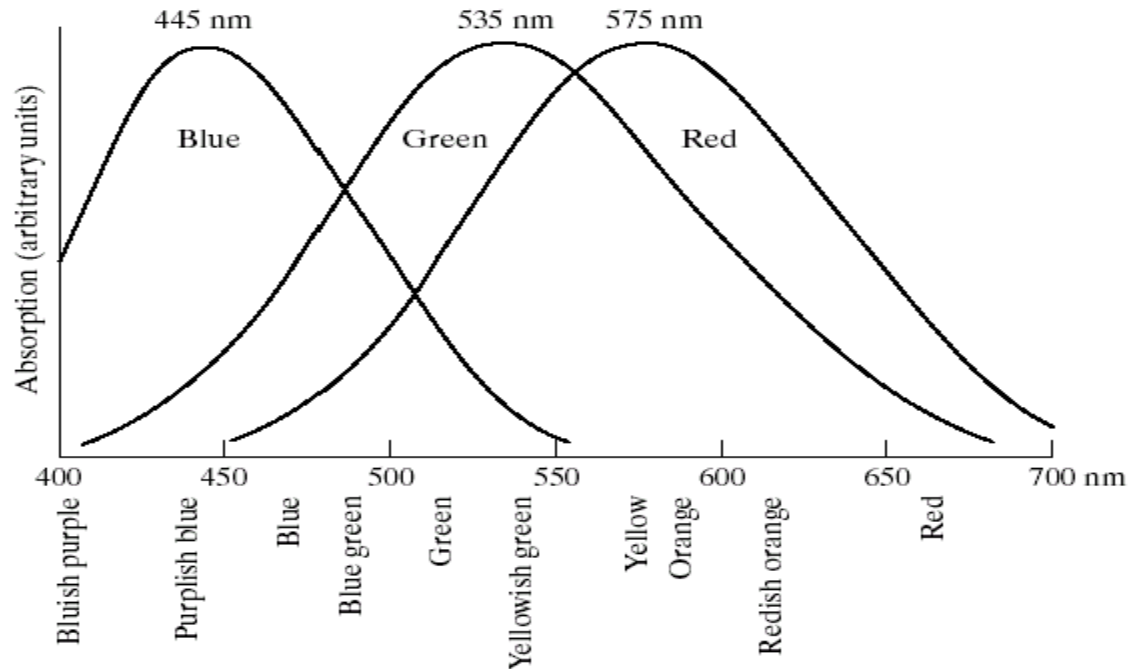
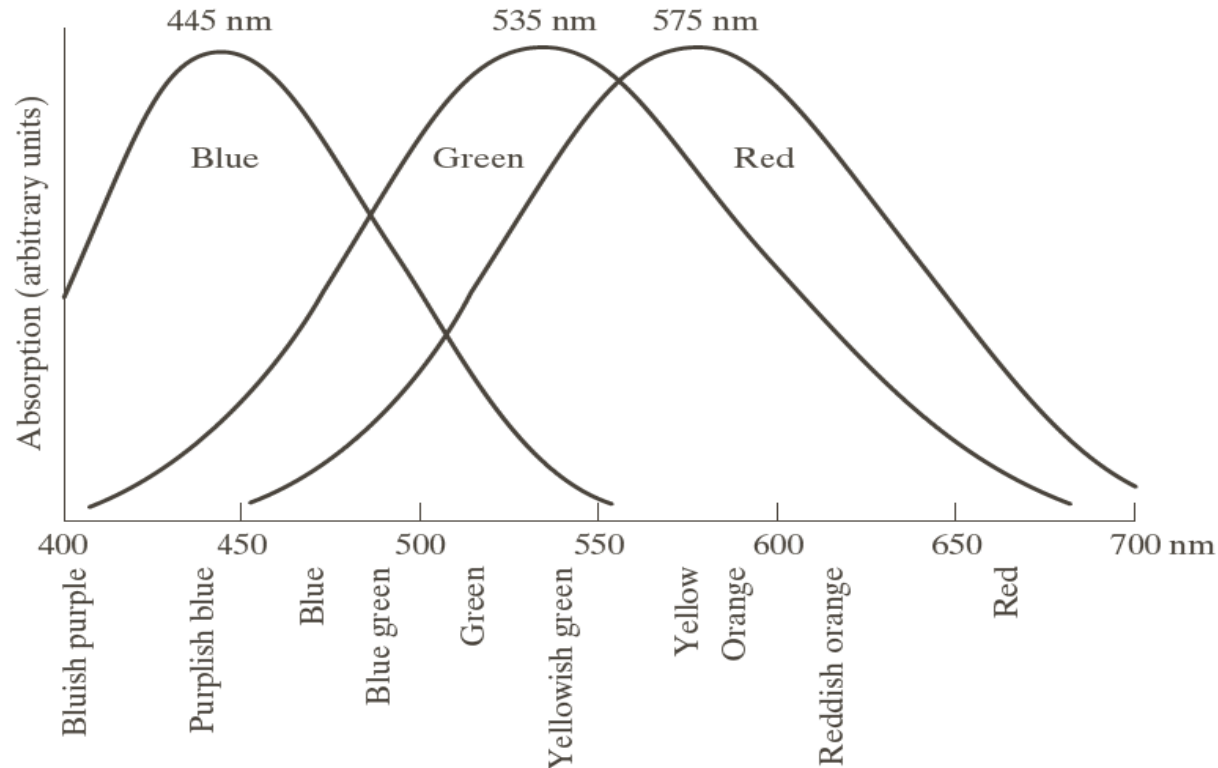


FIGURE 6.3 Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.

- ❑ Human eyes have three types of cone receptors that detect light in red, green, blue bands
- ❑ Mixing the three primaries or a secondary with its opposite primary colors in the right intensities produces white light

Color Fundamentals

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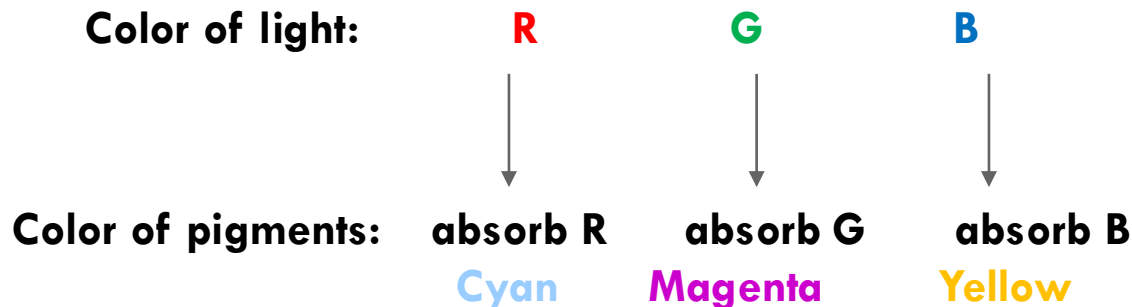


- ❑ A pure orange object will result in a different tristimulus response (different RGB values)
- ❑ We can create the perception of yellow with a combination of primary colors: $Y = R + G$

Primary Color of Light vs Primary Color of Pigments

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- ❑ Red, Green and Blue Colors are Primary Colors of Light
- ❑ In **Primary Color of Pigments**, a primary color is defined as the one that subtracts or absorbs a primary color of Light and reflects or transmits the other two
- ❑ Therefore, the **Primary Colors of Pigments** are **Magenta**, **Cyan** and **Yellow** and secondary colors are Red, Green and Blue
- ❑ A proper combination of three pigment primaries or a secondary with its opposite primary produces Black
- ❑ Color Television Reception is an example of the additive nature of Light Colors



Color Fundamentals

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- ❑ Tri-stimulus values: The amount of Red, Green and Blue needed to form any particular color
- ❑ Denoted by: X, Y and Z
- ❑ Tri-chromatic coefficient: used to specify colors

$$x = \frac{X}{X + Y + Z} \quad y = \frac{Y}{X + Y + Z} \quad z = \frac{Z}{X + Y + Z}$$

$$x + y + z = 1$$

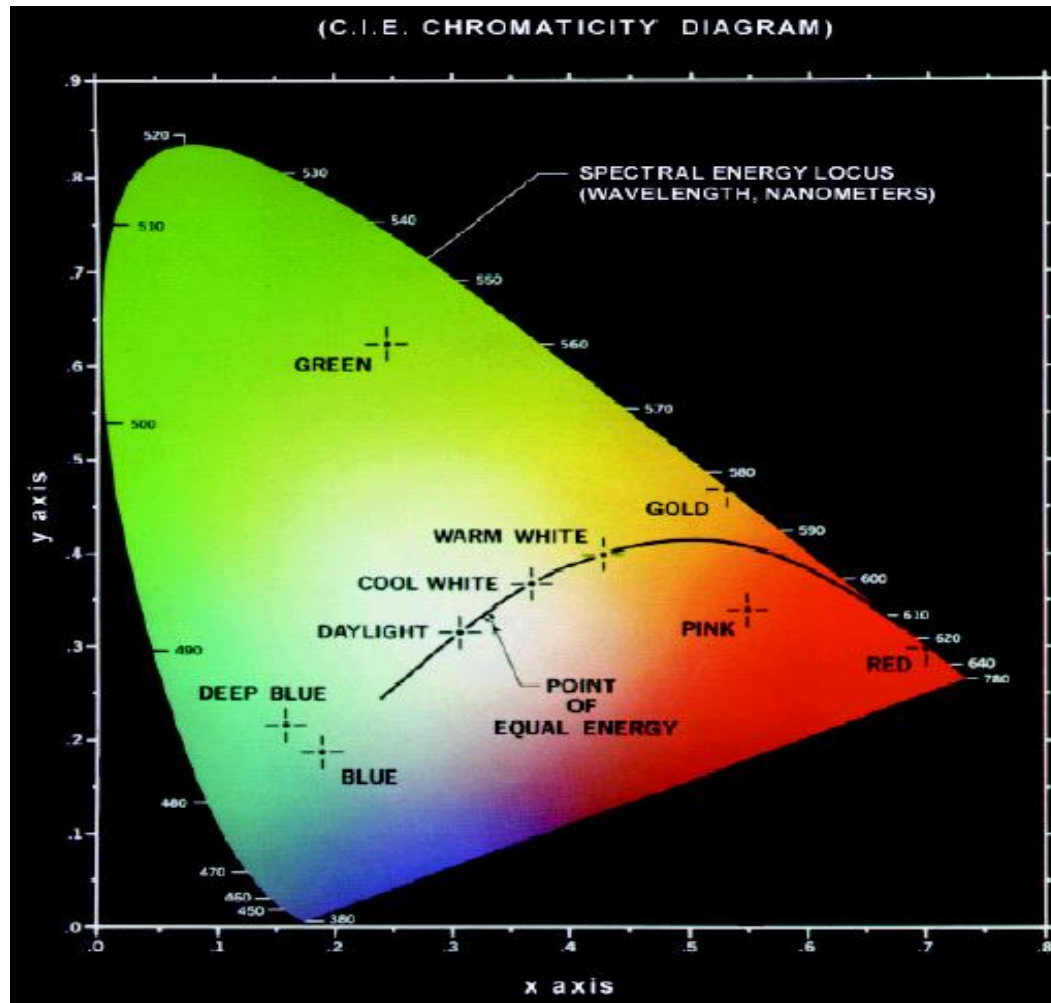
Chromaticity Diagram

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- ❑ CIE chromaticity diagram is normally used for specifying colors
- ❑ Shows color compositions as a function of x (red) and y (green)
- ❑ For any x and y , the corresponding value of z (blue) can be obtained as
$$z = 1 - x - y$$
- ❑ For example, the point marked green, in Chromaticity diagram Figure shown in next slide, has app. 62% green and 25% red content, so the composition of blue will be app 13%
- ❑ Chromaticity diagram is also useful for color mixing

Chromaticity Diagram

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Chromaticity Diagram

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- ❑ The positions of various spectrum colors are indicated around the boundary of the tongue shaped chromaticity diagram, these are the pure colors.
- ❑ Any point not actually on the boundary but within the diagram represents some mixture of spectrum colors
- ❑ The point of equal energy corresponds to equal fractions of the three standard primary colors; it represents the CIE standard of white light
- ❑ Any point on the boundary of the Chromaticity diagram is fully saturated
- ❑ As a point leaves the boundary and approaches the point of equal energy, more white light is added to the color, and it becomes less saturated
- ❑ **The saturation at the point of equal energy is equal to zero**

Chromaticity Diagram

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- ❑ Any straight line joining two points in the diagram defines all the different colors that can be obtained by combining these two colors additively
- ❑ This can be easily extended to three points
- ❑ The entire color range cannot be displayed based on any three colors

Chromaticity Diagram

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- ❑ To determine the range of colors that can be obtained from the 3 given colors in the CD, we simply draw connecting lines to each of the three-color points.
- ❑ The result is a triangle and any color inside a triangle is produced by various combinations of the three initial colors.
- ❑ The triangle shows a typical range of colors (called the color gamut) produced by RGB monitor

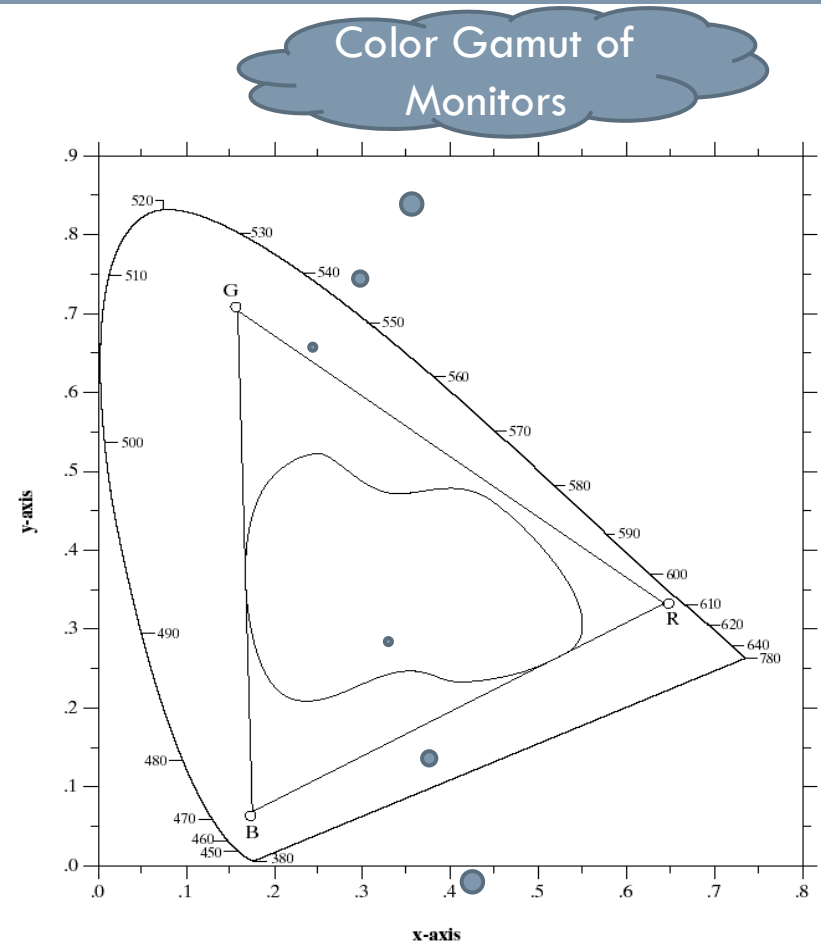


FIGURE 6.6 Typical color gamut of color monitors (triangle) and color printing devices (irregular region).

Color Models

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- ❑ The purpose of a color model (also called Color Space or Color System) is to facilitate the specification of colors in some standard way
- ❑ A color model is a specification of a coordinate system and a subspace within that system where each color is represented by a single point
- ❑ Some Color Models are listed below
 - ❑ **RGB**
 - ❑ **CMY**
 - ❑ **CMYK**
 - ❑ **YIQ**
 - ❑ **HSI**

} Suitable for hardware or applications

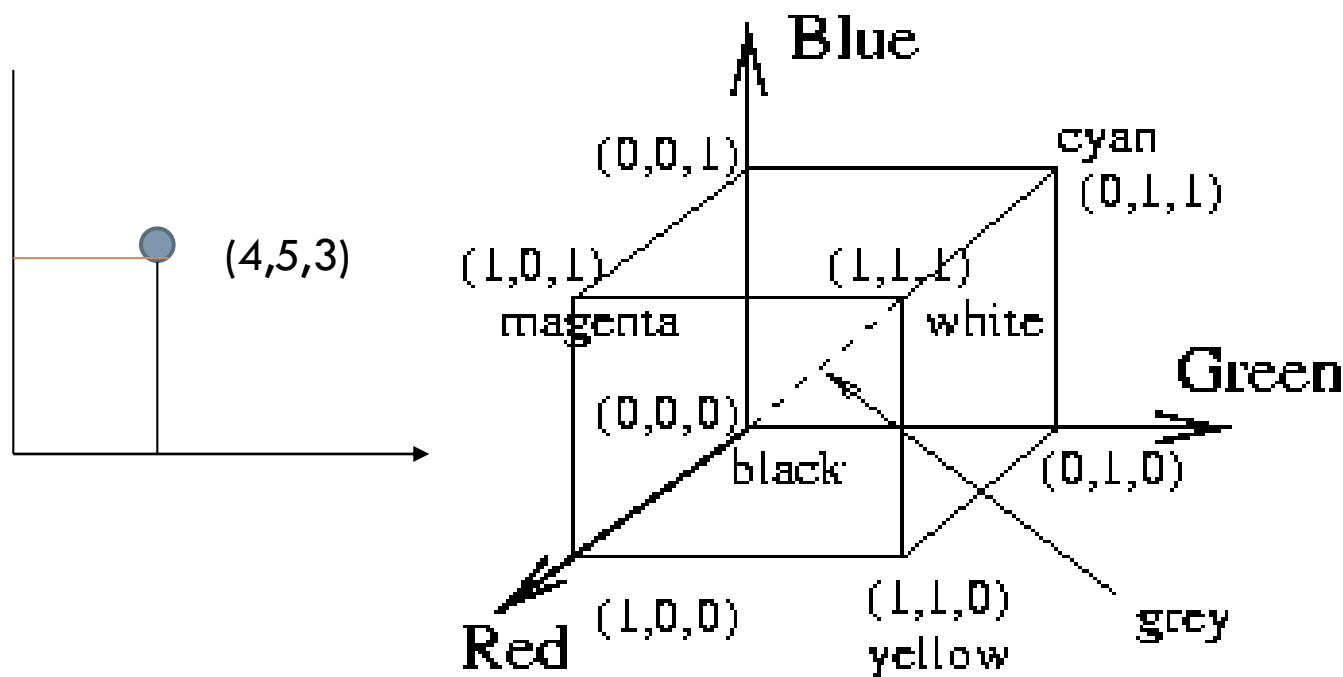
match the human description



RGB Model

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- Each color is represented in its primary color components **Red**, **Green** and **Blue**
- This model is based on **Cartesian Coordinate System**



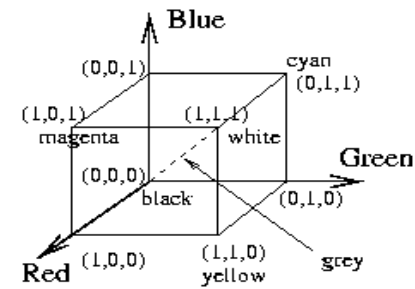


RGB Model

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- ❑ A **Cartesian coordinate system** is a coordinate system that specifies each point uniquely in a plane by a pair of numerical coordinates, which are the signed distances from the point to two fixed perpendicular directed lines, measured in the same unit of length.
- ❑ Each reference line is called a *coordinate axis* or just *axis* of the system, and the point where they meet is its *origin*, usually at ordered pair $(0, 0)$.
- ❑ The coordinates can also be defined as the positions of the perpendicular projections of the point onto the two axes, expressed as signed distances from the origin.
- ❑ One can use the same principle to specify the position of any point in three-dimensional space by three Cartesian coordinates.
- ❑ In general, n Cartesian coordinates specify the point in an n -dimensional Euclidean space for any dimension n .

RGB Model



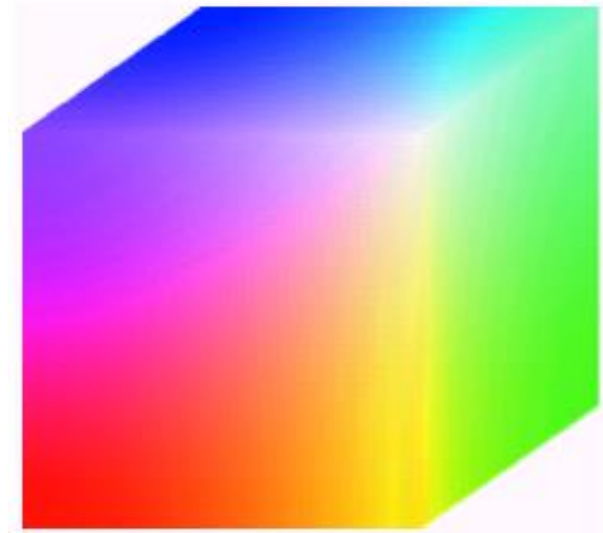
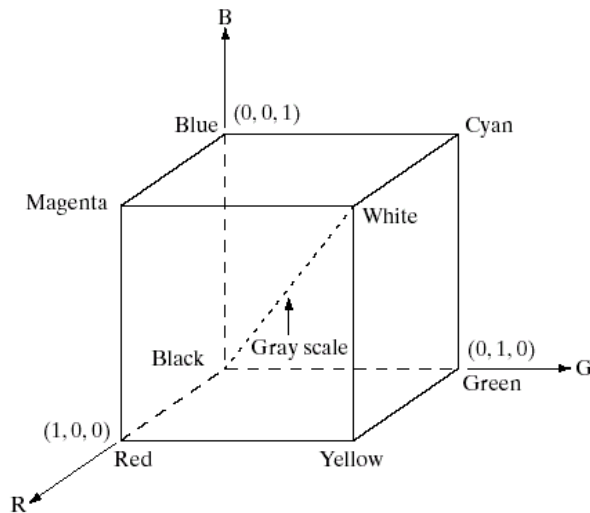
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- ❑ In this model, the primary colors are red, green, and blue. It is an additive model, in which colors are produced by adding components, with white having all colors present and black being the absence of any color.
- ❑ This is the model used for active displays such as television and computer screens.
- ❑ The RGB model is usually represented by a unit cube with one corner located at the origin of a three-dimensional color coordinate system, the axes being labeled R, G, B, and having a range of values $[0, 1]$.
 - ✓ RGB values are at 3 corners
 - ✓ Cyan magenta and yellow are at three other corners
 - ✓ Black is at the origin $[0,0,0]$
 - ✓ White is the corner furthest from the origin at $[1,1,1]$
 - ✓ The line joining black to white represents a gray scale and has equal components of R, G, B.
 - ✓ Different colors are points on or inside the cube represented by RGB vectors

Pixel Depth

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- ❑ **Pixel depth:** the number of **bits** used to represent each pixel in RGB space
- ❑ **Full-color** image: 24-bit RGB color image
 - $(R, G, B) = (8 \text{ bits}, 8 \text{ bits}, 8 \text{ bits})$



Safe RGB Color

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- ❑ Subset of colors is enough for some application
- ❑ Safe RGB colors (safe Web colors, safe browser colors)

Number System		Color Equivalents				
Hex	00	33	66	99	CC	FF
Decimal	0	51	102	153	204	255

TABLE 6.1

Valid values of each RGB component in a safe color.

$$(6)^3 = 216$$

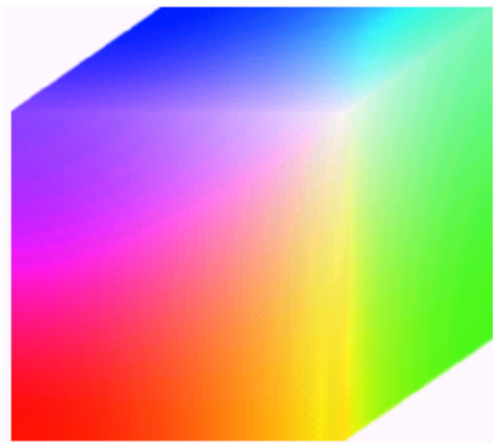


FIGURE 6.8 RGB 24-bit color cube.

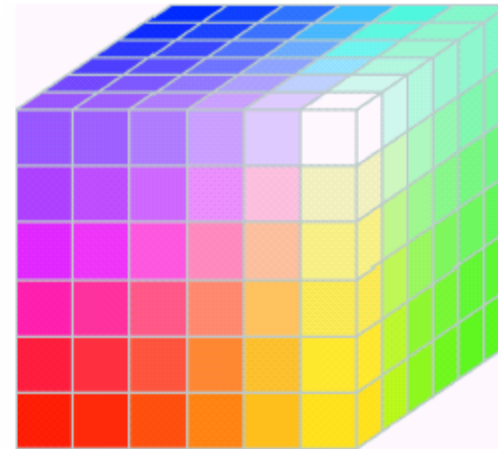


FIGURE 6.11 The RGB safe-color cube.

The total number of colors in a 24 Bit image is $(2^8)^3 = 16,777,216$ (> 16 million)

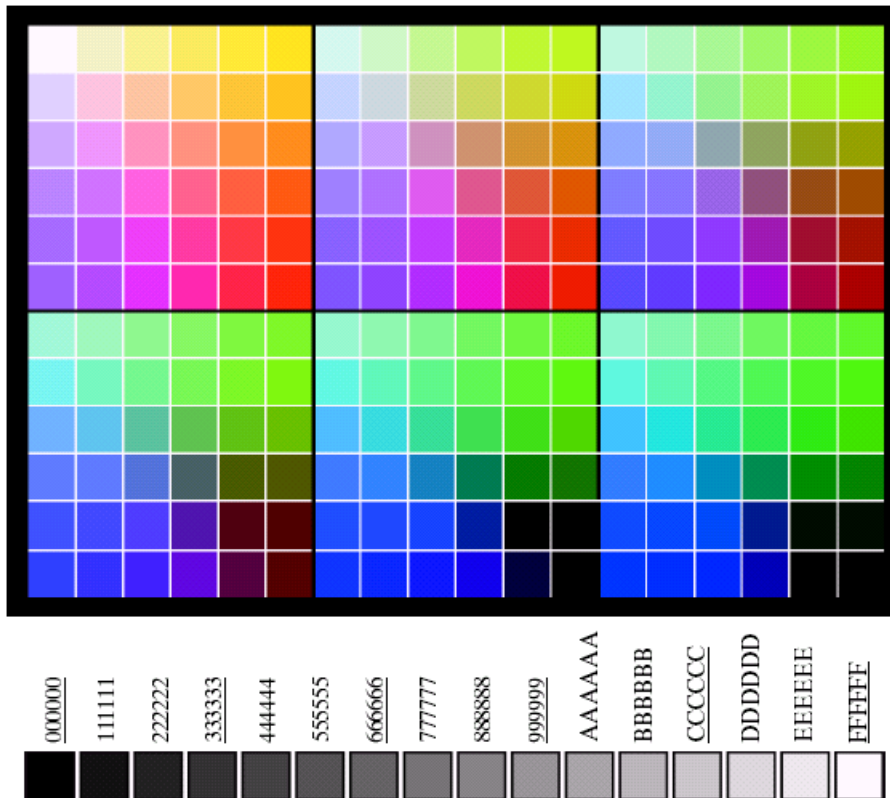
Safe RGB Color

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Number System	Color Equivalents					
Hex	00	33	66	99	CC	FF
Decimal	0	51	102	153	204	255

TABLE 6.1

Valid values of each RGB component in a safe color.



a
b

FIGURE 6.10

(a) The 216 safe RGB colors.
(b) All the grays in the 256-color RGB system (grays that are part of the safe color group are shown underlined).

CMY and CMYK Color Model

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- ❑ Cyan, magenta, and yellow are the secondary colors with respect to the primary colors of red, green, and blue. However, in this subtractive model, they are the primary colors and red, green, and blue, are the secondaries. In this model, colors are formed by subtraction, where adding different pigments causes various colors not to be reflected and thus not to be seen. Here, white is the absence of colors, and black is the sum of all of them. This is generally the model used for printing.
- ❑ Most devices that deposit color pigments on paper (such as Color Printers and Copiers) requires CMY data input or perform RGB to CMY conversion internally

$$\begin{pmatrix} C \\ M \\ Y \end{pmatrix} = \begin{pmatrix} 1.00 \\ 1.00 \\ 1.00 \end{pmatrix} - \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

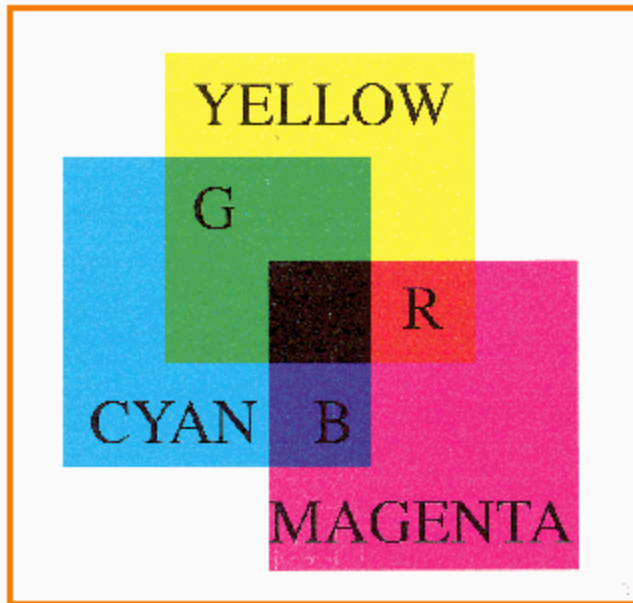
CMY and CMYK Color Model

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- ❑ CMY is a **Subtractive Color Model**
- ❑ Equal amounts of Pigment primaries (**Cyan, Magenta and Yellow**) should produce Black
- ❑ In practice combining these colors for printing produces a **“Muddy-Black”** color
- ❑ So, in order to produce **“True-Black”** a fourth color “Black” is added giving rise to CMYK model

CMY Color Model

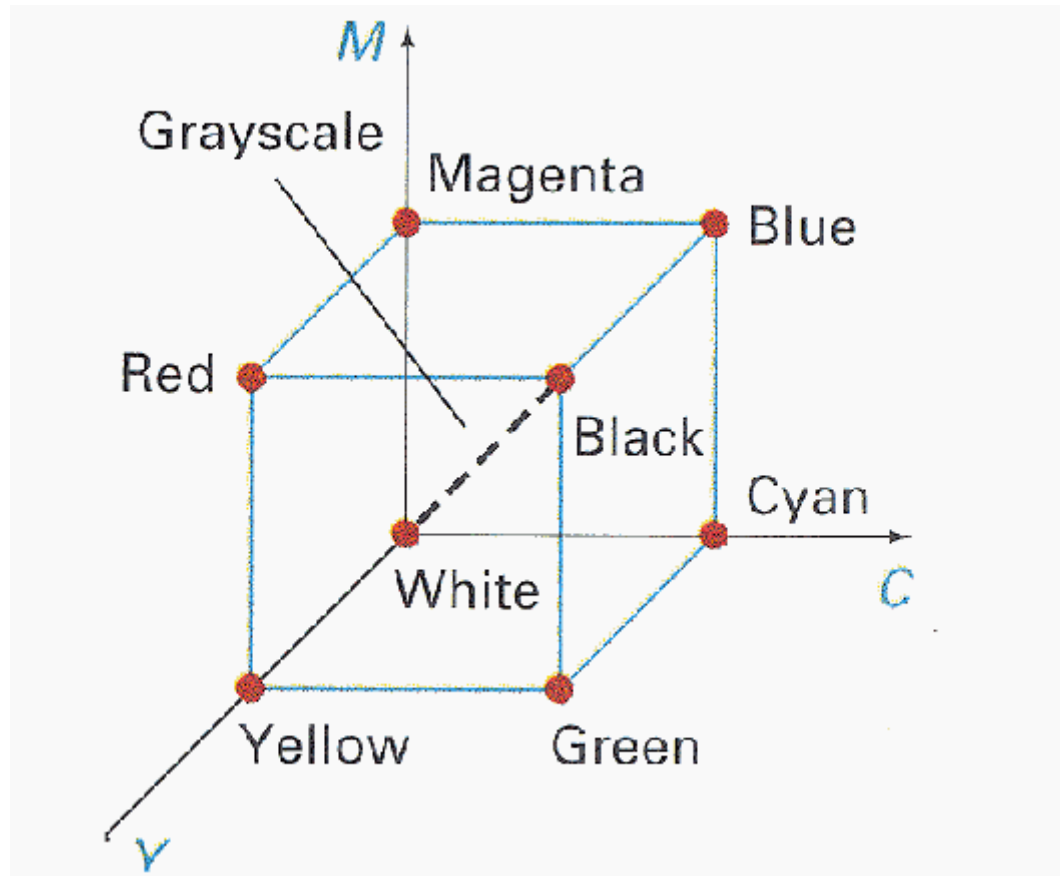
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C	M	Y	Color
0.0	0.0	0.0	White
1.0	0.0	0.0	Cyan
0.0	1.0	0.0	Magenta
0.0	0.0	1.0	Yellow
1.0	1.0	0.0	Blue
1.0	0.0	1.0	Green
0.0	1.0	1.0	Red
1.0	1.0	1.0	Black
0.5	0.0	0.0	
1.0	0.5	0.5	
1.0	0.5	0.0	

CMY Color Model

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HSI Model

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- ❑ RGB is useful for hardware implementations and is related to the way in which the human visual system works
- ❑ However, RGB is not a particularly natural way in which to describe colors
- ❑ Rather when people describe colors, they tend to use **hue**, **saturation** and **brightness**
- ❑ RGB is great for color generation, but HSI is great for color description

HSI Model

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□ Hue (dominant colour seen)

- ✓ Wavelength of the pure colour observed in the signal.
- ✓ Distinguishes red, yellow, green, etc.
- ✓ More than 400 hues can be seen by the human eye.

□ Saturation (degree of dilution)

- ✓ Inverse of the quantity of “white” present in the signal. A pure colour has 100% saturation, the white and grey have 0% saturation.
- ✓ Degree of Saturation: Inversely proportional to the amount of white light added
- ✓ Distinguishes red from pink, marine blue from royal blue, etc.
- ✓ About 20 saturation levels are visible per hue.

□ Intensity

- ✓ Distinguishes the gray levels.

HSI Model

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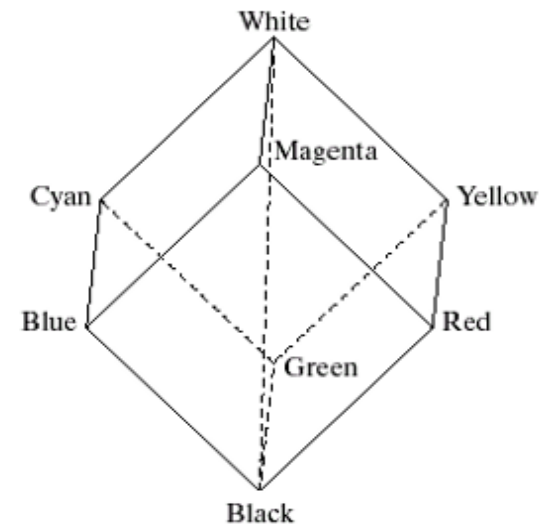
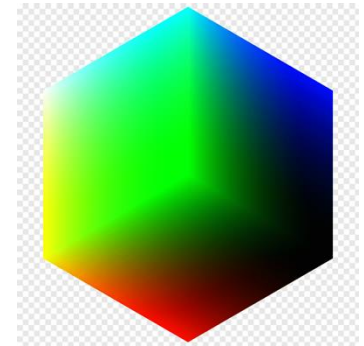
- ❑ Separates out intensity I from the image
- ❑ Two values (**Hue & Saturation**) encode *chromaticity*
- ❑ **Intensity** encode *monochrome part*.
- ❑ Hue and saturation of colors respond closely to the way humans perceive color, and thus this model is suited for interactive manipulation of color images.



HSI Model

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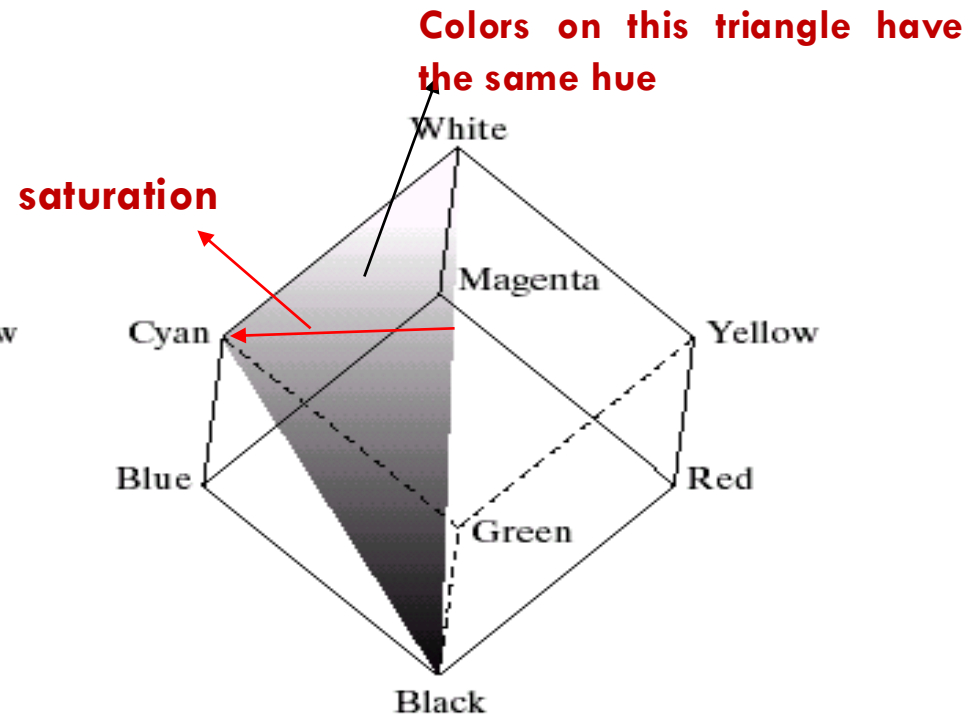
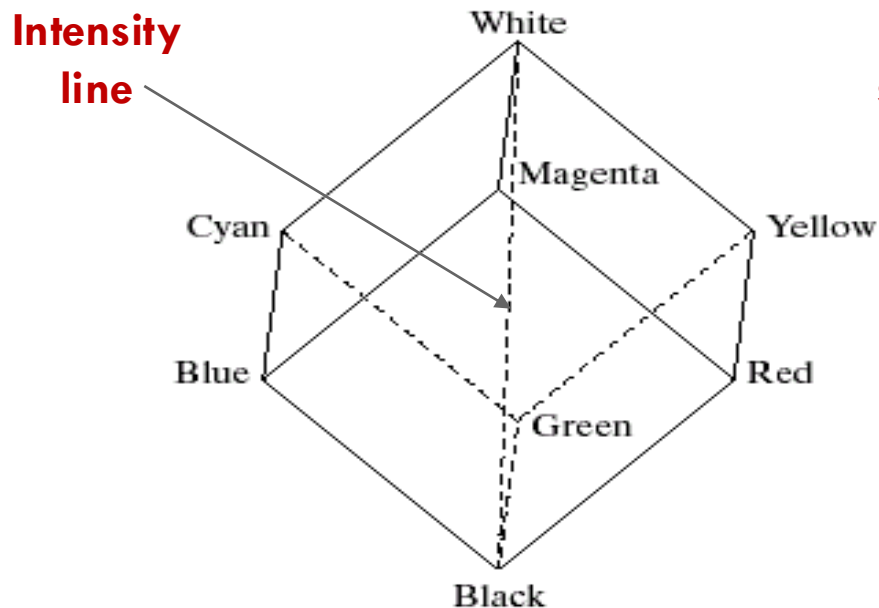
- Intensity can be extracted from RGB images – which is not surprising if we stop to think about it
- Remember the diagonal on the RGB color cube that we saw previously ran from black to white
- Now consider if we stand this cube on the black vertex and position the white vertex directly above it
- Now the intensity component of any color can be determined by passing a plane *perpendicular* to the intensity axis and containing the color point
- The intersection of the plane with the intensity axis gives us the intensity component of the color



HSI Model

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- In a similar way we can extract the hue from the RGB color cube
- Consider a plane defined by the three points cyan, black and white
- All points contained in this plane must have the same hue (cyan) as black and white cannot contribute hue information to a color



HSI Components

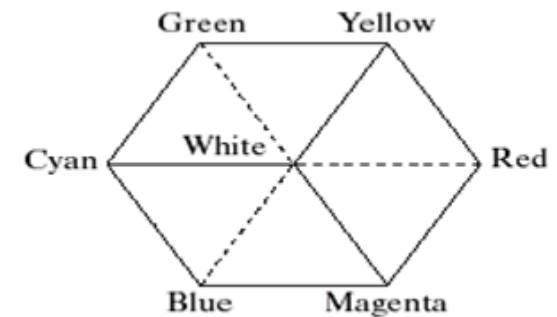
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HSI Model

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- ❑ Consider if we look straight down at the RGB cube as it was arranged previously
- ❑ We would see a hexagonal shape with each primary color separated by 120° and secondary colors at 60° from the primaries
- ❑ The hue is determined by an angle from a reference point, usually red
- ❑ The saturation is the distance from the origin to the point
- ❑ The intensity is determined by how far up the vertical intensity axis this hexagonal plane sits (not apparent from this diagram)



HSI Components

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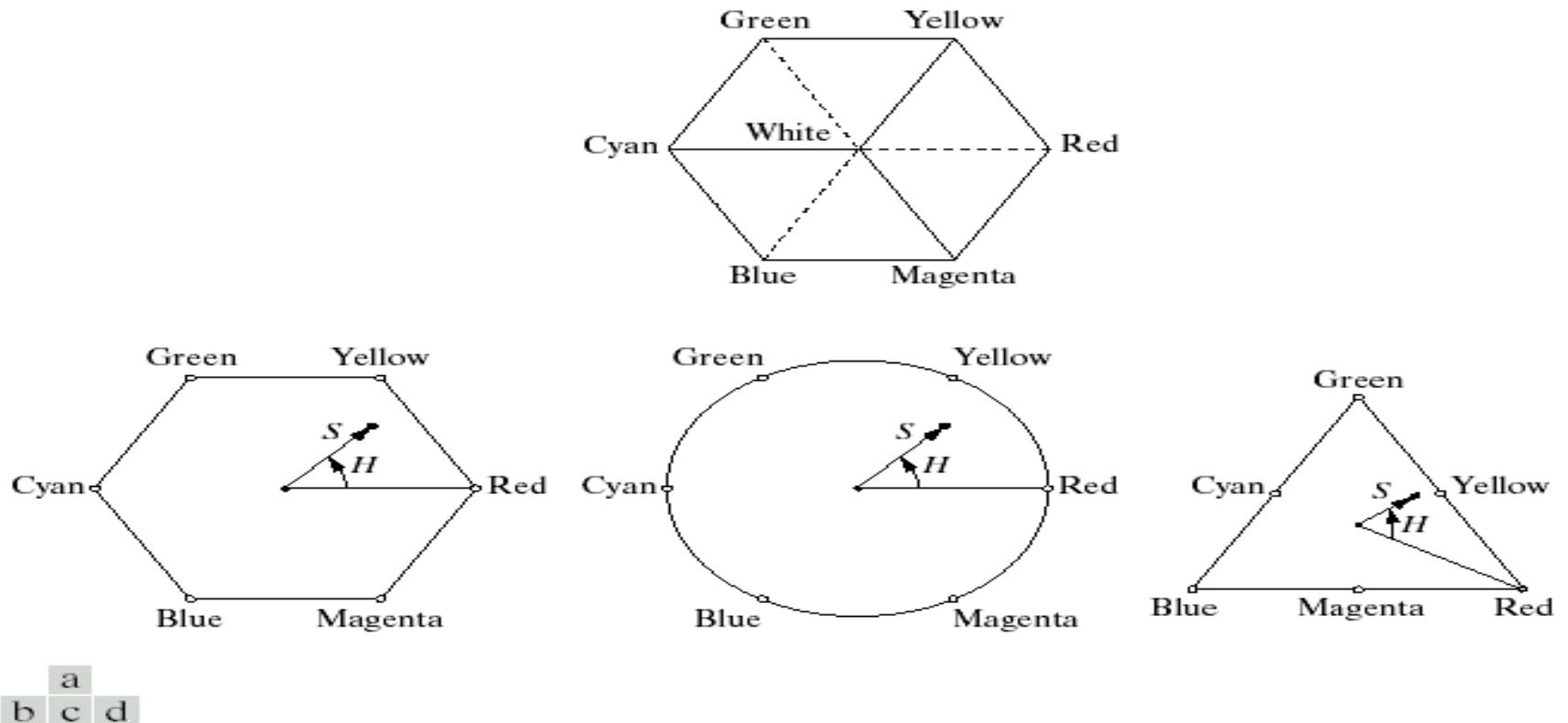
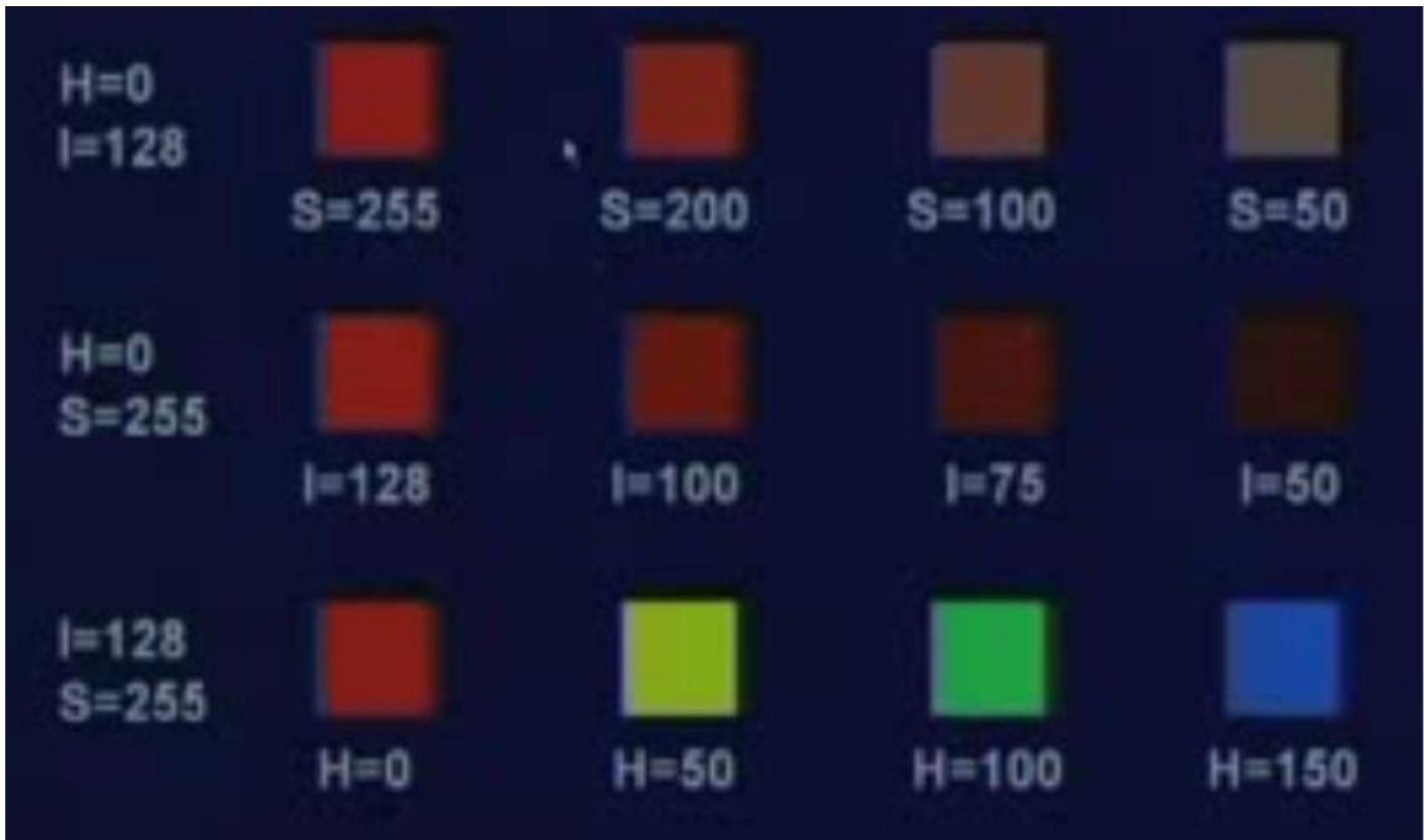


FIGURE 6.13 Hue and saturation in the HSI color model. The dot is an arbitrary color point. The angle from the red axis gives the hue, and the length of the vector is the saturation. The intensity of all colors in any of these planes is given by the position of the plane on the vertical intensity axis.

Variations of HSI values

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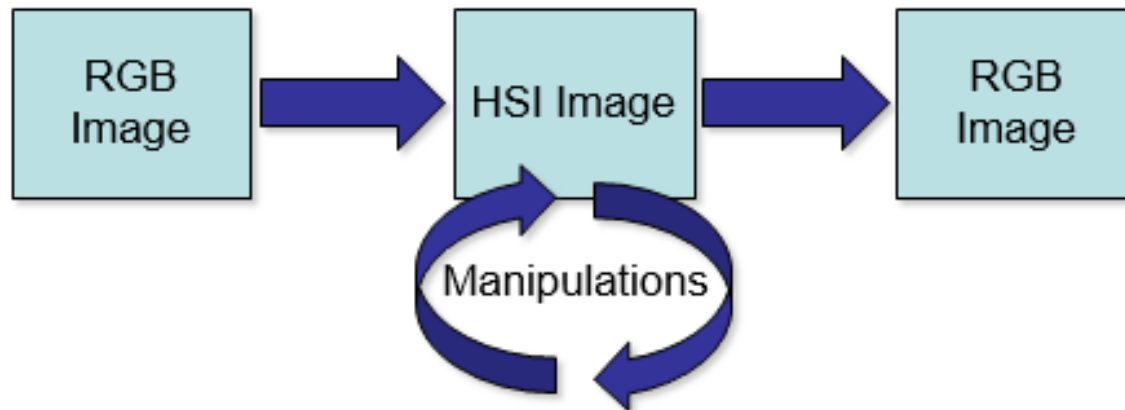


Manipulating Images in the HSI Model

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□ In order to manipulate an image under the HSI model we:

- ✓ First convert it from RGB to HSI
- ✓ Perform our manipulations under HSI
- ✓ Finally convert the image back from HSI to RGB



YIQ Model

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- ❑ The YIQ (luminance-inphase-quadrature) model is a recoding of RGB for color television and is a very important model for color image processing.
- ❑ Part of NTSC standard for color TV since 1953
- ❑ Used in North and South America and Japan
- ❑ Y is luma (only part shown on black/white TVs)
- ❑ I and Q represent chrominance, and are often down sampled to reduce image space/bandwidth

YIQ Model

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- The conversion from RGB to YIQ is given by:

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.275 & -0.321 \\ 0.212 & -0.523 & 0.311 \end{bmatrix} \times \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

- The conversion from YIQ to RGB is given by:

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 & 0.956 & 0.620 \\ 1 & -0.272 & -0.647 \\ 1 & -1.108 & 1.705 \end{bmatrix} \times \begin{bmatrix} Y \\ I \\ Q \end{bmatrix}$$

YUV Model

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- ❑ Part of PAL standard for color TV since 1963
- ❑ Used in Europe, Asia, Africa and Australia
- ❑ Y is luma (only part shown on black/white TVs)
- ❑ U and V represent chrominance, and are often down sampled to reduce image space/bandwidth

YUV Model

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- The conversion from RGB to YUV is given by:

$$\begin{bmatrix} Y \\ U \\ V \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.147 & -0.289 & 0.436 \\ 0.615 & -0.515 & -0.100 \end{bmatrix} \cdot \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

- The conversion from YUV to RGB is given by:

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1.140 \\ 1 & -0.395 & -0.580 \\ 1 & 2.032 & 0 \end{bmatrix} \cdot \begin{bmatrix} Y \\ U \\ V \end{bmatrix}$$

References

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1. DIP by Gonzalez

For any query Feel Free to ask



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